5

Classification and Supervised Pattern Recognition

5.1 Introduction

5.1.1 Background

Pattern recognition has been the fastest growing application of chemometrics in the past decade, especially in the application to metabolomics. The origins of chemometrics were primarily in analytical chemistry, with chemical engineering (primarily in the area of process control) also taking an important front seat in the 1980s and 1990s.

However, during the past few years, there has been substantial increase in applications primarily to biology and medicine, where the aim is not so much to measure an analyte to a high degree of accuracy, or determine kinetics or equilibria, or identify components of a mixture but to use laboratory-based data, primarily from spectroscopy or chromatography, to determine the provenance of a sample.

Most of these problems can be formulated as classification problems. That is, samples are assigned using their analytical signal into one or several predefined classes, or groups, or labels. Pattern recognition, as originally defined in the 1960s, was quite broadly based, for example, involving handwriting analysis and facial recognition; however, over the past few days, it has primarily involved classification. There are two main forms of pattern recognition, unsupervised, as discussed in Chapter 4, and supervised, the basis of this chapter.

The original applications of supervised pattern recognition in chemometrics were primarily concerned with classifying samples into two or more known classes. In classical analytical chemistry, the reference standards were known in advance; for example, we might know that an extract came from one of the two types of orange juice, one from Brazil and one from Greece. We are certain of the provenance of the orange, and we may be developing a method, for example, in near infrared (NIR) to assign the spectra into one of the classes. The higher the classification ability, the better the method. There is a huge and somewhat misleading literature in which algorithms are compared according to their classification ability.

Some of this classical literature is not directly applicable to many modern-day problems.

- We may not be certain that the analytical technique can separate the groups; for example, can we use the HPLC method in urine samples to unambiguously determine whether patients are diseased or not; the aim may be to hypothesise that groups can be separated; thus, the aim of pattern recognition is hypothesis testing.
- There may be additional groups that we do not know about, or subgroups, or mislabelled samples.
- There are often outliers, or atypical samples, for example, originating from patients with unusual genetics that are not typical of the background population.
- We may be interested in markers, or variables.

Some people use simulations to test the relative merits of algorithms, but this depends on the simulations being realistic – the characteristics of a population are rarely known perfectly in advance. However, simulations can be used to develop and understand how methods work.

Every method has its merits. A lot depends on the aim of the study, which is known in advance, either with certainty or hypothesised. In this chapter, we will first look at some of the main methods and then how to assess them. We will not discuss Bayesian enhancements (see Section 3.6.3) in this chapter. They are often used in the machine-learning community but much less common in the chemometrics community.

Chemometrics: Data Driven Extraction for Science, Second Edition. Richard G. Brereton.
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Table 5.1 Case study in Section 5.1.2: the data involve 20 samples in two classes (first 10 = class A, second 10 = class B) recorded using two variables.

Sample	x_1	x_2
1	0.847	1.322
2	0.929	0.977
3	1.020	1.547
4	0.956	0.842
5	1.045	1.687
6	1.059	1.363
7	0.860	2.876
8	0.973	2.383
9	0.680	0.201
10	1.171	1.964
11	1.428	1.226
12	1.372	0.982
13	1.118	0.616
14	0.900	0.266
15	1.766	1.453
16	1.273	0.580
17	1.298	0.929
18	1.478	1.018
19	1.388	0.858
20	1.081	0.584

5.1.2 Case Study

In order to illustrate the methods of this chapter, we will describe a simple case study. This involves 20 samples, 10 from one class (A) and 10 from another class (B), recorded using two variables. These data are presented in Table 5.1 and illustrated in Figure 5.1.

1. The matrix in Table 5.1 has dimensions
   (a) 2×20
   (b) 20×2
   (c) $10 \times 2 \times 2$
   (d) $2 \times 10 \times 2$

5.2 Two-Class Classifiers

Two-class classifiers aim to find a boundary between two groups. Samples at one side of the boundary are assigned to one group (which we will call A) and the other side to the other group (which we will call B). This is illustrated in Figure 5.2(a).

The boundary can be of varying complexity. Different classifiers are defined according to the nature of this boundary, as discussed below, and it does not have to be linear. There is no requirement that this boundary perfectly separates the classes, as shown in Figure 5.2(b). If there, for example, is no linear boundary, either the classifier should be changed or the data are really inseparable using commonly available methods.

Two-class classifiers force all samples to be assigned to one of the two groups and do not usually allow for outliers or ambiguous samples.

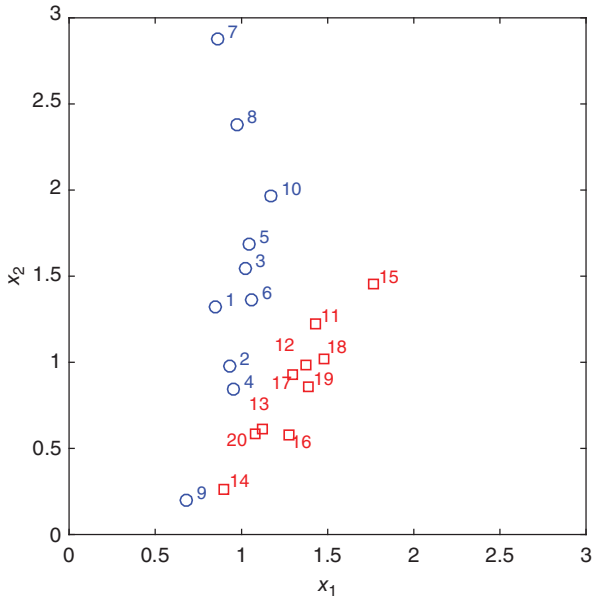


Figure 5.1 Data set in Table 5.1.

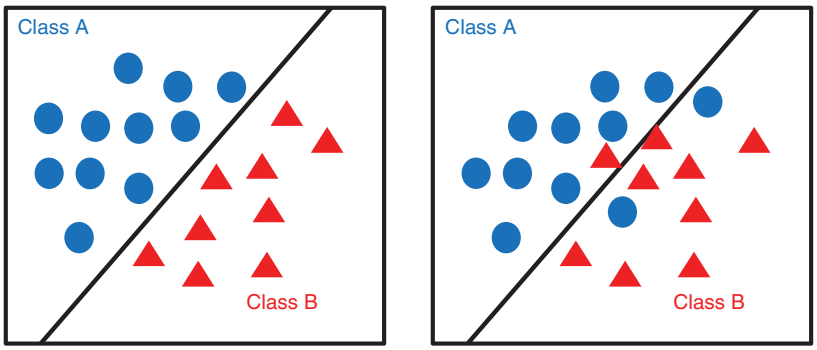


Figure 5.2 Two-class classifiers. (a) Linearly separable classes. (b) Linear inseparable classes.

1. There are 20 samples of polluted soils and 20 of unpolluted soils. A linear two-class classifier is constructed and classifies 18 of the polluted and 16 of the unpolluted correctly.
   (a) Two polluted are misclassified as unpolluted and four unpolluted as polluted.
   (b) Four polluted are misclassified as unpolluted and two unpolluted as polluted.
   (c) Six samples are misclassified, but we cannot assign these into any specific group.

5.2.1 Distance-Based Methods

Distance-based methods are conceptually the simplest and based on traditional statistics.

- For each group, the centroid is calculated. This is the average of all the samples in the group.
- Then, for each sample in the group, a distance to the centroid of each group is computed. We will discuss the different distance measures below.
- In the simplest situation, sample is assigned to the group whose centroid it is closest to.

In our example, the centroid of group A is $\begin{bmatrix}0.955 & 1.517\end{bmatrix}$ and group B is $\begin{bmatrix}1.311 & 0.852\end{bmatrix}$. Of course, the data can be described by any number of variables. Hence, the centroid of a group of dimensions 50×400, which might represent

5.2.1.1 Euclidean Distance to Centroids

The simplest type of distance is the Euclidean distance defined in vector terminology by

$$d_{i\mathrm{A}}^2 = (\boldsymbol{x}_{i\mathrm{A}} - \bar{\boldsymbol{x}}_{\mathrm{A}})(\boldsymbol{x}_{i\mathrm{A}} - \bar{\boldsymbol{x}}_{\mathrm{A}})'$$

where

- A is the class and i the sample,
- $\boldsymbol{x}_{i\mathrm{A}}$ is the row vector corresponding to the measurements for this sample and
- $\bar{\boldsymbol{x}}_{\mathrm{A}}$ is the centroid of this class.

In our case, the square Euclidean distance from class A for sample 6 is calculated as follows.

- $\bar{\boldsymbol{x}}_{\mathrm{A}}$ is [0.955 1.517] or the average of the first 10 readings.
- Hence, $\boldsymbol{x}_{6\mathrm{A}} - \bar{\boldsymbol{x}}_{\mathrm{A}} = [1.060 \quad 1.364] - [0.955 \quad 1.517] = [0.105 \quad -0.153]$.
- So $d_{6\mathrm{A}}^2 = (\boldsymbol{x}_{6\mathrm{A}} - \bar{\boldsymbol{x}}_{\mathrm{A}})(\boldsymbol{x}_{6\mathrm{A}} - \bar{\boldsymbol{x}}_{\mathrm{A}})'$ can be calculated by the sum of squares of $\boldsymbol{x}_{6\mathrm{A}} - \bar{\boldsymbol{x}}_{\mathrm{A}}$ or $0.105^2 + (-0.153)^2 = 0.0345$.
- Therefore, the distance of sample 6 to the centroid of class A is the square root of 0.0345, or 0.186.
- Note that the units will be in the original measurement units. Sometimes, of course, the two variables measured will be in different units (e.g. pH and temperature), in which case either it is a combination or variables are transformed to similar units by standardisation (Section 4.6); however, more usually they are the same, for example, absorbances at different wavelengths.

The distance of this sample to class B is 0.754. Hence, it is closer to the centroid of class A and assigned to this class. This is the principle of the method of Euclidean distance to centroids (EDC).

1. The centroid of a group is [0.78 0.92 1.70 −1.33 0.68]. An individual sample has measurements [0.54 0.23 0.68 0.91 1.32]. What is the Euclidean distance of the sample to the centroid?
   (a) 0.93
   (b) 0.96
   (c) 7.00
   (d) 2.65

5.2.1.2 Linear Discriminant Analysis

However, different variables may be on quite different scales, consider one variable being a weight in kilogram and the other a length in metre. Should we use the same scale for each variable or weight them so that they are of equal significance, or variance? Another problem is that sometimes some variables are correlated. For example, we might be analysing the spectra of a set of compounds, and there could be one broad peak corresponding to several wavelengths, all varying in the same way, and they could unfairly weight the analysis.

To correct for this, we often calculate the Mahalanobis distance. The most straightforward method is called linear discriminant analysis (LDA). This involves weighting the variables by the inverse of the pooled variance–covariance matrix – so

$$d_{i\mathrm{A}}^2 = (\boldsymbol{x}_{i\mathrm{A}} - \bar{\boldsymbol{x}}_{\mathrm{A}})\boldsymbol{S}_p^{-1}(\boldsymbol{x}_{i\mathrm{A}} - \bar{\boldsymbol{x}}_{\mathrm{A}})'$$

(the variance and covariance are described in Section A.1). The pooled variance–covariance matrix is the average over both groups defined by

$$\boldsymbol{S}_p = [(I_{\mathrm{A}} - 1)\boldsymbol{S}_{\mathrm{A}} + (I_{\mathrm{B}} - 1)\boldsymbol{S}_{\mathrm{B}}]/(I_{\mathrm{A}} + I_{\mathrm{B}} - 2)$$

where I_{A} is the number of samples in group A (in our case 10) and $\boldsymbol{S}_{\mathrm{A}}$ its variance–covariance matrix (using the population variance) and similarly for group B. If both groups have equal number of samples, then it is simply the average of the variance–covariance matrices. Of course, if there are more than two groups, an equivalent equation can be found with as many groups as necessary.

For calculations in, for example, Excel or Matlab, you should use the population variance or covariance because we are primarily scaling variables rather than estimating population parameters.

- For our data set,

$$\boldsymbol{S}_{\mathrm{A}} = \begin{bmatrix} 0.0167 & 0.0381 \\ 0.0381 & 0.5367 \end{bmatrix} \quad \text{and} \quad \boldsymbol{S}_{\mathrm{B}} = \begin{bmatrix} 0.0519 & 0.0711 \\ 0.0711 & 0.1096 \end{bmatrix}$$

(c) $\begin{bmatrix} 0.2821 & 0.1192 & 0.1366 \\ 0.1192 & 0.1904 & 0.0595 \\ 0.1366 & 0.0595 & 0.2821 \end{bmatrix}$

5.2.1.3 Quadratic Discriminant Analysis

LDA uses the pooled variance–covariance matrix of an entire data set, consisting of, in our example, two classes. It assumes that the variance of both\classes is the same. In some cases, this is an adequate assumption, and the main issue is that the variables themselves are correlated or on different scales.

However, sometimes each group has its own structure. Consider trying to separate elephants from mice, the variance of the elephants is very much larger than that of the mice. The result is that small elephants could be mistaken as large mice unless we modelled each group separately. To do this, a simple adjustment is involved. Instead of using the overall variance, calculate the variance of each group separately so that

$$d_{i\mathrm{A}}^2 = (\boldsymbol{x}_{i\mathrm{A}} - \bar{\boldsymbol{x}}_{\mathrm{A}})\boldsymbol{S}_{\mathrm{A}}^{-1}(\boldsymbol{x}_{i\mathrm{A}} - \bar{\boldsymbol{x}}_{\mathrm{A}})'$$

where $\boldsymbol{S}_{\mathrm{A}}$ is the variance–covariance matrix of group A with a similar equation for group B, which is the basis of quadratic discriminant analysis (QDA).

- In the example in Table 5.1,

$$\boldsymbol{S}_{\mathrm{A}} = \begin{bmatrix} 0.0167 & 0.0381 \\ 0.0381 & 0.5367 \end{bmatrix} \quad \text{and} \quad \boldsymbol{S}_{\mathrm{B}} = \begin{bmatrix} 0.0519 & 0.0711 \\ 0.0711 & 0.1096 \end{bmatrix}$$

  as discussed above.
- Hence,

$$\boldsymbol{S}_{\mathrm{A}}^{-1} = \begin{bmatrix} 71.370 & -5.061 \\ -5.061 & 2.222 \end{bmatrix}$$

- This means for the sixth sample, the distance from group A is given by

$$\begin{aligned} d_{6\mathrm{A}}^2 &= \begin{bmatrix} 0.105 & -0.153 \end{bmatrix} \begin{bmatrix} 71.370 & -5.061 \\ -5.061 & 2.222 \end{bmatrix} \begin{bmatrix} 0.105 \\ -0.153 \end{bmatrix} \\ &= \begin{bmatrix} 8.269 & -0.872 \end{bmatrix} \begin{bmatrix} 0.105 \\ -0.153 \end{bmatrix} = 8.292 \times 0.105 + (-0.875 \times -0.153) \\ &= 1.002 \end{aligned}$$

- Hence, the distance is the square root of 1.002, or 1.001, Mahalanobis distance units from the centre of group A.

QDA allows each group to have its own variance structure. In our example, the correlation between the two variables for the overall data set is −0.027; hence, they are virtually uncorrelated. If we consider just group A alone, it is 0.401, and group B alone 0.942; thus, each group on its own has quite different correlations. This is reflected in the variance–covariance matrix $\boldsymbol{S}$ for the overall data set where the off-diagonal elements are relatively small, compared with $\boldsymbol{S}_{\mathrm{B}}$ where they are relatively large.

As usual, the number of variables should not exceed the number of samples in the smallest group. Although this can be overcome, as for LDA, PCA would have to be on each group separately and is rarely done for two-class classifiers. One-class classifiers overcome this and are described in the next section.

1. A data set consists of 50 samples, 20 in group A and 30 in group B, and seven variables. The dimensions of the variance–covariance matrix of group A are
   (a) 20×20
   (b) 2×2
   (c) 7×7

2. LDA uses the pooled variance–covariance matrix for the full data set, where there are two or more groups, in contrast to QDA, which uses the variance–covariance matrix of each group separately.
   (a) True
   (b) False

5.2.1.4 Comparing EDC, LDA and QDA

In order to understand the difference between the three methods, we compute the distances to the centroid of all 20 samples in Table 5.1. These are presented in Table 5.2. Samples are assigned to the class they are nearest to. For example, using QDA, we find that sample 6 has a distance of 1.001 to group A but 7.793 to class B; therefore, it is assigned to class A.

In our example, we find that there are three misclassified samples using EDC and one using QDA. All are correctly classified using LDA.

In order to better understand this, it is useful to visualise the boundaries between the classes graphically, as shown in Figure 5.4. The boundaries are the points where $d_A = d_B$ or the distance to the centroids of each class is equal, samples on either side of this are assigned into one of the classes. We also draw the equidistance contours for each class. We can see that for EDC, these are circular (or if there are more than two variables, hyper-spherical). For LDA, they form ellipses, the direction of the ellipses being given by the maximum variance of the pooled variance–covariance matrix: we see that for both classes, they are in the same direction as a 'common' matrix is computed. For QDA, they are in different directions, reflecting the different structure and orientation of the groups.

In addition, we see that the QDA boundaries are curved (or in our case parabolic). In some situations, we do not expect two groups to be linearly separable and as such these are more realistic boundaries. For the case study in this section, it is reasonable to use a linear boundary as LDA performs slightly better to QDA, but in other cases, groups are not linearly separable. It is also important to note that for the methods in this section, we just look at which side of the boundary a sample lies, but it is of course possible to look at how close a sample is to a boundary and form a probabilistic model, as discussed in Section 5.3.

Table 5.2 Class distances for the data in Table 5.1 using EDC, LDA and QDA together with the predicted class memberships.

Sample	True class	EDC			LDA			QDA		
		d_{iA}	d_{iB}	Predicted	d_{iA}	d_{iB}	Predicted	d_{iA}	d_{iB}	Predicted
1	A	0.222	0.660	A	0.580	3.525	A	0.831	10.167	A
2	A	0.540	0.401	B	1.036	2.550	A	0.744	6.075	A
3	A	0.073	0.754	A	0.388	2.850	A	0.541	9.936	A
4	A	0.674	0.354	B	1.393	2.226	A	1.012	4.563	A
5	A	0.194	0.877	A	0.494	2.957	A	0.706	10.870	A
6	A	0.186	0.570	A	0.869	2.314	A	1.001	7.793	A
7	A	1.363	2.074	A	3.146	6.139	A	2.456	23.890	A
8	A	0.867	1.568	A	1.724	4.627	A	1.236	18.031	A
9	A	1.343	0.905	B	2.337	3.479	A	2.357	3.367	A
10	A	0.498	1.121	A	1.191	2.845	A	1.680	11.765	A
11	B	0.556	0.393	B	3.341	0.744	B	4.197	1.992	B
12	B	0.678	0.145	B	3.343	0.340	B	3.920	0.497	B
13	B	0.915	0.304	B	2.547	1.048	B	2.284	0.881	B
14	B	1.251	0.715	B	2.412	2.219	B	1.731	1.813	A
15	B	0.814	0.755	B	5.195	2.474	B	6.898	2.011	B
16	B	0.989	0.274	B	3.433	0.480	B	3.498	1.992	B
17	B	0.681	0.079	B	2.982	0.211	B	3.355	0.854	B
18	B	0.723	0.237	B	3.938	0.929	B	4.774	0.930	B
19	B	0.788	0.078	B	3.631	0.484	B	4.159	0.962	B
20	B	0.941	0.352	B	2.432	1.254	B	2.069	1.091	B

Of course, the converse is not true for the sum of squares to another group, or out-group, as any fresh sample could be an outlying sample and will have no influence on the centroid or variance–covariance matrix of the in-group.

As an example, the sum of squares of the 10 samples of group A for the QDA distances to the centroid of group B is 1484.30.

Note that this property is not valid for LDA because the pooled variance–covariance matrix depends on both groups, whereas the centroid is unique to each group. For the purpose of brevity, we do not expand on these properties in this text.

1. A data set consists of 50 samples, 20 in group A and 30 in group B, and three variables. The sum of squares of the Mahalanobis distance using QDA for samples in group A to their centroid is
   (a) 20
   (b) 60
   (c) 3
   (d) 150
2. QDA will always classify more samples correctly to LDA as boundaries are quadratic and so extra terms can be added to better define the boundary.
   (a) True
   (b) False

5.2.2 Partial Least-Squares Discriminant Analysis

PLS-DA has become very topical over the past decade. In Chapter 6 and Section A.2.2, we discuss the algorithm that will not be reproduced here. The description in this section will exclusively relate to a two-class problem and PLS1.

The basis of PLS-DA is that the classifier is given a numerical value c for each sample, often +1 for class A and −1 for class B, and the $\boldsymbol{X}$ matrix is calibrated against this. Using a model

$$\hat{c} = \boldsymbol{tq}$$

for each sample (see Section 6.5.1 for more details), an estimated value of c can be obtained. This predicts the class membership for each sample. In the simplest case, if it is >0, it is usually assigned to class A, otherwise assigned to class B. This is called a decision threshold, and in the case of, for example, unequal class sizes of differing variances, the decision threshold can be changed, but the simplest and default criterion is used below.

It is usual to centre $\boldsymbol{X}$ before performing PLS and compute the scores using centred $\boldsymbol{X}$. For each sample, the vector $\boldsymbol{t}$ will be of dimensions $1 \times A$, where A is the number of PLS components retained and $\boldsymbol{q}$ will be of dimensions $A \times 1$. Usually, the $\boldsymbol{X}$ matrix is centred down the columns before performing PLS (as discussed in Chapter 6); however, if there are unequal class sizes, it is better to adjust the $\boldsymbol{X}$ matrix by the average of the means of the two classes so that

$$\boldsymbol{x}_{inew} = \boldsymbol{x}_i - (\bar{\boldsymbol{x}}_{i\mathrm{A}} + \bar{\boldsymbol{x}}_{i\mathrm{B}})/2$$

where $\mathbf{x}_{\boldsymbol{inew}}$ is the transformed data for sample $\boldsymbol{i}$ and $\bar{\boldsymbol{x}}_{i\mathrm{A}}$ the mean or centroid of class A. In our example, both classes have equal sizes; thus, this transformation is the same as column mean centring.

- In our example, if we keep both PLS components, $\boldsymbol{q}$ is a vector $\begin{bmatrix}1.061\\1.984\end{bmatrix}$.
- The PLS scores of both components of the centred data are given in Table 5.3.
- Hence, for sample 6, we have $\hat{c} = \begin{bmatrix}0.193 & 0.061\end{bmatrix}\begin{bmatrix}1.061\\1.984\end{bmatrix} = 0.326$.
- If we keep only one component, it is $0.193 \times 1.061 = 0.204$.
- In both cases, the predicted value is >0 and the sample is assigned to class A according to both criteria.

The predicted values $\hat{c}$ are presented in Table 5.4. We can then assign the samples into their predicted groups, usually according to whether $\hat{c}$ is positive or negative. Note an important conclusion:

- If all non-zero PLS components are used, the position of the PLS boundary and assignment to classes is identical to that using LDA.

Table 5.4 PLS-DA predictions of c for one-component and two-component models for the centred data in Table 5.4.

Sample	True class	One component	Predicted	Two components	Predicted
1	A	0.272	A	0.856	A
2	A	−0.092	B	0.385	A
3	A	0.396	A	0.565	A
4	A	−0.231	B	0.214	A
5	A	0.514	A	0.602	A
6	A	0.204	A	0.326	A
7	A	1.718	A	1.969	A
8	A	1.201	A	1.306	A
9	A	−0.692	B	0.471	A
10	A	0.710	A	0.473	A
11	B	−0.109	B	−0.752	B
12	B	−0.309	B	−0.784	B
13	B	−0.524	B	−0.382	B
14	B	−0.742	B	−0.063	B
15	B	−0.066	B	−1.478	B
16	B	−0.635	B	−0.818	B
17	B	−0.321	B	−0.627	B
18	B	−0.328	B	−1.037	B
19	B	−0.433	B	−0.917	B
20	B	−0.535	B	−0.308	B

example using modern data-rich instrumentation, as there is no need to calculate the Mahalanobis distance (although performing PCA before the distance calculations is also possible).

1. A data set consists of 200 samples, 100 from class A and 100 from class B; there are 50 variables. The data matrix is centred and PLS-DA performed.
   (a) A model with one PLS component will result in the same predictions as LDA.
   (b) A model with 50 PLS components will result in the same prediction as LDA.
   (c) We cannot say whether the same prediction ability is obtained as from LDA no matter how many components, the methods are unrelated.

5.2.3 K Nearest Neighbours

One of the oldest classifiers in chemometrics is k nearest neighbours (kNN). The basis of the method is simple.

- Choose k. This is usually an odd number from 3 onwards.
- Calculate the distance of the sample of interest from all the samples in the training set. Usually, the Euclidean distance is used. This should include itself (the reasons will be discussed below).
- Select the k samples that are closest.
- By a 'majority vote', assign the sample to the most popular class. For example, if $k = 5$, and the nearest samples (including itself) are members of classes A, A, B, A, A, then by a majority of 4 to 1, it is assigned to class A.
- There are various rules in the case of ties. When there are only two classes and k is an odd number, there will be no ambiguity.
- Note that in the algorithm described above, the closest sample will always be itself.

We list the five nearest samples for the data set in Table 5.1 and the assignments using $k = 3$ and $k = 5$ in Table 5.5. We see that when $k = 3$, only sample 9 is misclassified, whereas when $k = 5$, both samples 4 and 9 are misclassified.

5.3 One-Class Classifiers

In the case of two-class classifiers, we determined what is often called a hard boundary between two classes. We insist that each sample is unambiguously assigned to one or other of the classes according to which side of the boundary it falls. Another type of classifier is called a one-class classifier. The question asked is not 'Is a sample a member of either class A or B?' but 'Is a sample a member of class A or not?'.

This is equivalent to drawing a boundary around a class, as illustrated in the left-hand diagram of Figure 5.7, rather than between classes. The advantage of this is that each class can be modelled separately. In addition, more, new, classes can be considered without changing the original models. There are also several additional features of one-class classifiers. The first is that it is possible to have ambiguous samples, as they may appear to be a member of both classes simultaneously. This could occur because the analytical method is insufficient to distinguish between the classes or because it genuinely has both characteristics, an example being a compound that is both a ketone and an ester, it may have spectral characteristics of both types of functional group. The second is that there can be outliers, which are members of no prior class, they may be a mislabelled sample or genuinely from a new class. Hence, if we are modelling two classes, there can be four verdicts rather that two, for a two-class problem, namely class A, class B, ambiguous or outlier.

1. We are trying to model three classes. Using one-class classifiers, how many possible verdicts are there?
   - **(a)** Three types of unambiguous samples, three types of ambiguous samples, three types of outliers.
   - **(b)** Three types of unambiguous samples, four types of ambiguous samples, three types of outliers.
   - **(c)** Three types of unambiguous samples, four types of ambiguous samples, one type of outliers.
   - **(d)** Three types of unambiguous samples, three types of ambiguous samples, one type of outliers.

5.3.1 Quadratic Discriminant Analysis

The simplest and most straightforward method is QDA. This method can be used both as a two-class classifier (Section 5.2.1.3) or a one-class classifier. In its usual form, it is based on the assumption that the data are (multi)-normally distributed. As a two-class classifier, a sample is assigned to the class it is closest to. As a one-class classifier, a sample is assigned to a class if it is within a defined distance (often converted to a probability) from the centroid.

Figure 5.8 illustrates a typical data set in which the samples are multi-normally distributed, which is defined as follows.

- Each variable (in our case, two variables) in itself must be distributed normally.
- The sum of any combination of variables must also be distributed normally. In algebraic terms, if variable 1 is distributed normally, we call this x_1, then $ax_1 + bx_2 + \cdots + nx_j$ must also form a normal distribution, where a, b and so on can be any numbers.

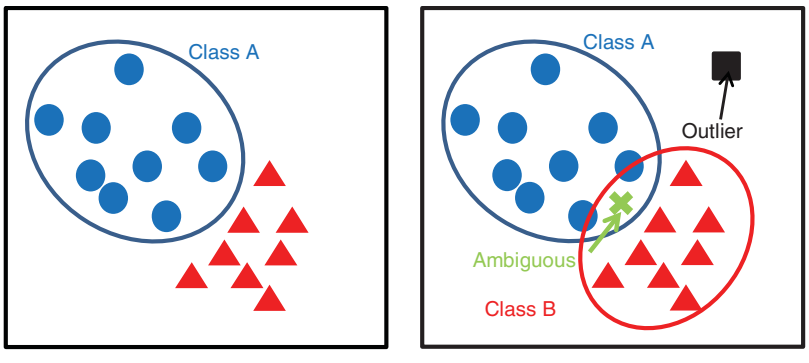


Figure 5.7 One-class classifiers: (a) separable class; (b) classes with ambiguous and outlying samples.

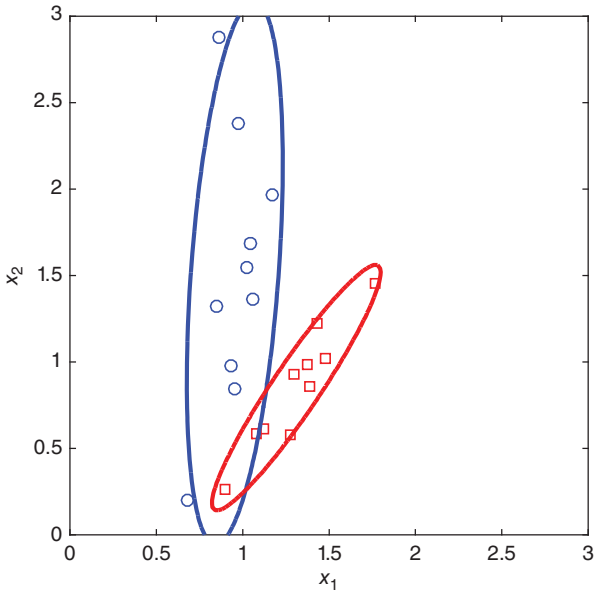


Figure 5.9 QDA one-class boundaries using 90% confidence ($p = 0.1$) for data in Table 5.1.

Some papers, instead of using a χ^2 criterion, use the Hotelling's T^2 as an alternative. This is related to the F distribution (see Sections A.3.4 and A.3.5), with a scaling factor, and the interested reader is referred to the detailed literature. Hotelling's T^2 is designed for very small sample sizes; however, both measures also assume underlying normality, which is rarely perfectly the case; hence, once there are more than half a dozen samples, the difference is only, in practice, useful for checking simulations.

It is also important to remember that the Mahalanobis distance cannot be calculated if there are more variables than samples, unless PCA is performed first on the data, as discussed in Section 5.2.1.2.

1. A data set is characterised by 10 variables; what is the Mahalanobis distance corresponding to a value of $p = 0.05$ assuming a χ^2 distribution?
   (a) 18.31
   (b) 16.92
   (c) 4.11
   (d) 4.28
2. A sample is at three Mahalanobis distance units from the centroid of class A and 2.5 from the centroid of class B. Four variables have been measured. If we use a 95% confidence limit ($p = 0.05$), using a χ^2 distribution,
   (a) The sample is assigned to class B.
   (b) The sample is assigned to neither class.
   (c) The sample is ambiguous.

5.3.2 Disjoint PCA and SIMCA

Self-independent modelling of class analogy (SIMCA) was probably the first method reported, in the 1970s, to be specifically aimed at chemometricians. It would be recognised nowadays as a one-class classifier.

When the number of variables is large, it often makes sense to reduce the variables by PCA first, before classification, as discussed in Chapter 4, and perform the classification (e.g. QDA) on the reduced PC model. Traditionally, if we perform PCA, we would take the entire data set consisting of all classes and reduce the variables using PCA, which is called a conjoint model. However, there is a conceptual problem. If the aim is to do one-class classification on group A, which we call the *in-group*, should we take group B into account as well? If PCA is performed on all the data together, in practice, group B is influencing group A in the PC model. There is also another disadvantage. One-class classification should be performed independently on each class; thus, it should not matter how many classes there are, and if new

The advantage of disjoint models is that new classes can always be added to the data without disturbing the existing models. Another advantage is that PCA can be performed independently on any class, reducing the number of variables, which is often important in modern applications such as spectroscopy or chromatography where hundreds or indeed thousands of variables can be measured quite rapidly, and the principles in this section can easily be extended to multivariate situations. Of course, when there are several classes, it is not necessary to choose the same number of PCs for each class as they are independently modelled: in our case, the only way we can reduce the dimensionality is by going from two to one PC, but that is not of course the case when there are more than two variables; thus, for example, class A may be characterised by three PCs and class B by five PCs (in an imaginary example).

1. A data set consists of samples originating from three known classes.
   (a) We can form one type of conjoint PC model and one type of disjoint PC model.
   (b) We can form three types of conjoint PC models and three types of disjoint PC models.
   (c) We can form one type of conjoint PC model and three types of disjoint PC models.
   (d) We can form three types of conjoint PC models and one type of disjoint PC model.

5.4 Multi-Class Classifiers

When there are more than two classes, we have to adapt the methods discussed in Section 5.2 if we want to form hard boundaries between these classes, as illustrated in Figure 5.12.

For some types of classifiers, this procedure is quite straightforward. For distance-based methods discussed in Section 5.2.1, we simply take the distance (scaled as appropriate) to the nearest centroid. For kNN methods discussed in Section 5.2.3, assign to the class to which a sample has the most neighbours, with rules if there are ties (e.g. equally assign to more than one class).

However, for PLS-DA, the situation is significantly more difficult.

The simplest is to perform PLS-DA separately for each class using an approach called 'one versus all'. As we discuss in the next chapter, if there is only one c variable, we use the PLS1 algorithm by default. The c value is +1 for the in-group or class to be modelled and −1 for all the rest or out-group. Often, the $\boldsymbol{X}$ value is transformed separately for each calculation because the relative group sizes will differ, as discussed in Section 5.2.2. Hence, if there are three classes, three separate models are formed, namely (a) class A against all others, (b) class B against all others and (c) class C against all others. This is illustrated in Figure 5.13.

For each sample and each model, we will get an estimated value of c. For example, we may have three classes, and the estimated value of c for the class A model is −0.86, class B model +1.34 and class C model −1.20; therefore, it is assigned to class B. One problem, however, with these 'one versus all' models, is that there can frequently be ambiguity. For example, what happens if a sample has a positive estimated value of c for both class A and B models, and negative for class C, which do we assign it to? There are numerous rules, one might be that the sample is assigned to the class for which c is most positive, another for which it is closest to +1 – also remember that we could have a negative estimated value of c for all classes. There is no universal agreement or accepted rule, and there are often further complications if class sizes are unequal.

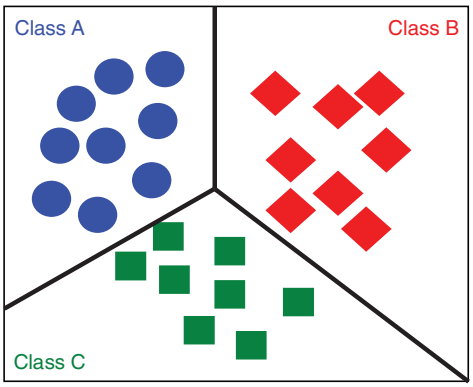


Figure 5.12 Multi-class classifiers.

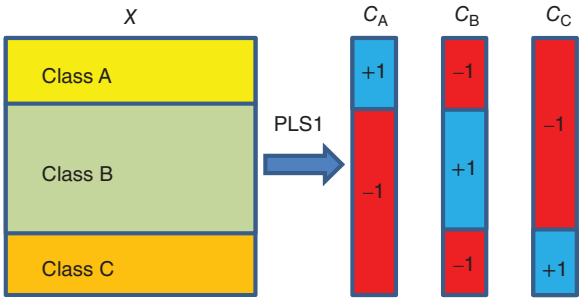


Figure 5.13 PLS1 multi-class models.

Some people instead use PLS2 (see Chapter 6), which models all classes at the same time. The $\boldsymbol{C}$ matrix consists of A columns, where A is the number of classes, each column corresponding to the c vector using PLS1. We will not discuss this method in detail, except to say that there are significant difficulties with transforming the $\boldsymbol{X}$ matrix, and often PLS2 results in quite misleading and hard to interpret results, hence is best avoided.

Many people using PLS-DA as a multi-class method are unaware of how the decisions are taken; for example, how columns are transformed, what is the decision criterion and whether PLS1 or PLS2 is used. For obvious groupings with clearly distinguished samples, this probably does not matter, but when there are tricky problems; for example, with overlapping groups, different group sizes, ambiguous or outlying samples, the method used is very critical to the model performance. The recommendation is not to use PLS-DA when there are more than two classes unless the problem is very straightforward or one is well aware of the steps in the algorithm employed. It is normally safer to use more straightforward approaches such as distance measures or one-class classifiers, which can of course be extended to any number of classes without difficulty.

1. PLS2 is always used as an alternative to PLS1 when there are more than two classes in a model for classification models.
   (a) True
   (b) False

5.5 Optimisation and Validation

Optimisation and validation are often confused. In the early days of chemometrics, and indeed multivariate classification literature, usually there was a known answer in advance. For example, there is classical work by the statistician R.A. Fisher who took 150 irises, divided into three classes each of difference species. There was no ambiguity about the desired answer. Hence, methods could be compared or optimised according to how well they predicted the known samples; that is, how well they were classified.

There is a huge chemometrics literature devoted to comparing and optimising classification methods. Investigators choose one or more favoured data sets and claim that method A (usually the one developed by the authors often in order to get a PhD or a grant) is better than all other methods. They often try hard to optimise their method, for example, by choosing an appropriate number of components (PCs or PLS components), tuning parameters (e.g. choosing the ones that give an answer that appears to lead to the highest correct classification rate), selecting variables and scaling and transforming data. An author may claim that their method results in 97% correctly classified, whereas a rival's method results in only 92%.

This mindset, in part, has its origins in analytical chemistry, where optimisation is an important aspect of most published methodological work. For example, chromatographic optimisation involves improving separations. Many laboratories report their performances against known reference standards, and the lower the error, the better the laboratory. In addition, much of traditional chemometrics also involved developing methods that tried to predict known answers – for example, to determine an NIR method that can predict HPLC quantification as accurately as possible and replace the definitive but slow HPLC method by a faster NIR technique. In all such cases, there is a well-established reference, and it is easy to say whether a chemometric method performs well or not.

However, chemometrics is now being employed in many other areas, for example, metabolomics or the study of human culture. In such situations, we may not be certain of the provenance of a sample even in the training set. As

an example, are we certain that a patient has a known disease; hence, do we expect the GCMS of a urine sample to perfectly classify patients into two groups, diseased or healthy? Indeed, the healthy patients may have other diseases, or unusual genetics, and there may be a progression in disease (severe or mild) or misdiagnosis. Or the analytical technique may not be perfect. In mainstream statistics, Bayesian methods (see Chapter 3) are often used under such circumstances, but the majority of chemometrics literature and software prefers other approaches, and certainly, the vast majority of literature comparing and announcing new chemometrics techniques is primarily based on simple performance criteria – in our case, usually which method correctly classifies more samples into predefined categories. The problem here is that when doing pattern recognition, we are also often testing hypotheses; for example, does the NMR of serum extracts tell us whether a disease has progressed or not? We do not know the answer in advance, and it is actually very hard to find a fully controlled and representative set of samples. Therefore, a method that, for example, classifies the maximum proportion of samples into our preconceived groups may not be the most appropriate and may suffer from what is called overfitting.

Hence, we have a problem as are often trying to do two quite separate things. The first is optimisation; that is, get as high a success rate as possible, but the second is validation, that is test the underlying hypothesis, for example, that there really is a separation between groups. Traditionally, these two rather different aims have been mixed up and methods such as cross-validation (discussed in Chapter 4) have been used for both purposes. If we are certain of the underlying hypothesis as may happen in traditional analytical chemistry (e.g. we may want to separate unadulterated from adulterated orange juices and do some experiments in the laboratory we are really sure of before analysing the extracts), this is quite legitimate. But in other cases, where we cannot be sure of the provenance or representativeness of samples, we need to separate validation from optimisation.

1. Extracts of urine are obtained from 100 subjects, 50 of which have a disease and 50 of which are controls (do not have the disease). Method A classifies 87% correctly into one of the two categories, whereas Method B classifies 94% correctly.
   (a) Method B is definitely more appropriate and hence superior.
   (b) We cannot tell.

5.5.1 Validation

5.5.1.1 Test Sets

This is the process of determining whether a hypothesis is correct. The usual approach is to divide the overall data (often called auto-predictive) into two parts, a training set and a test set. Typically, 2/3 of the samples are assigned to the training set, although other splits are possible. This is illustrated in Figure 5.14.

The model is developed and optimised using the training set, but model is validated on the test set. Consider a simple illustration of a seating plan. We might be sociologists and want to study whether people of different genders sit next to each other. For example, for an audience in a family entertainment, there may be more or less random distribution between males and females. However, when giving a more formal presentation to an audience in a traditional society

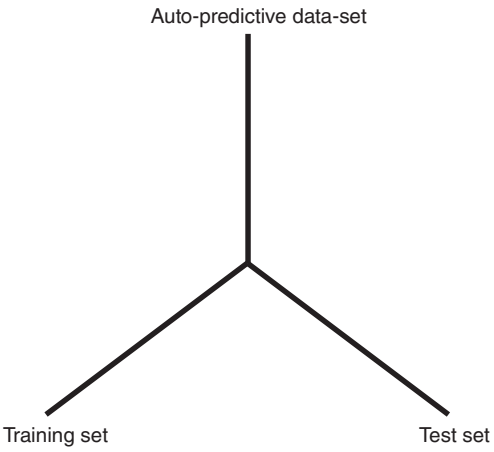


Figure 5.14 Division of data into training and test sets.

Table 5.10 EDC model of data in Table 5.1 divided into training and test sets.

Sample	Training set			Test set			
	d_{iA}	d_{iB}			d_{iA}	d_{iB}	
4	0.775	0.357	B	1	0.317	0.690	A
5	0.108	0.914	A	2	0.640	0.417	B
6	0.271	0.606	A	3	0.090	0.790	A
7	1.264	2.112	A	11	0.607	0.429	B
8	0.766	1.607	A	12	0.755	0.180	B
9	1.444	0.879	B	13	1.012	0.276	B
10	0.405	1.160	A				
14	1.352	0.684	B				
15	0.819	0.785	B				
16	1.082	0.236	B				
17	0.765	0.117	B				
18	0.789	0.264	B				
19	0.869	0.089	B				
20	1.039	0.325	B				

- As these values are very similar, we deduce that the model was not particularly overfitted (otherwise the training set %CC (per cent correctly classified) would be much higher than the test set) and probably there is reasonable evidence of separation between the groups.
- If we calculate the average Euclidean distance from the centroids over the training and test sets, it is $\begin{bmatrix}0.758 & 0.646\end{bmatrix}$ compared with $\begin{bmatrix}0.715 & 0.631\end{bmatrix}$ for the overall (auto-predictive) result in Table 5.2, which is to be expected.
- Note that for this example, the average distance of the test set samples to the training set centroid is in fact less than that for the training set because we chose only three samples and would differ according to the samples chosen; if all the three possible combinations of samples were selected, then the average distance should be larger. Hence, there is some importance to how samples are selected for the test set.

In practice, in our example, the assumption that there is a differentiation between the classes is a reasonable one, although there are one or two samples in a region of overlap. Obviously, there will be other examples, and other methods, where this will not be so; however, for brevity, we are mainly illustrating the principles of the methods rather than searching out specific case studies. However, validation can protect against overfitting and allow us to determine whether we are justified in using our model or not.

1. The proportion of samples correctly classified using a test set is normally less than that for the training set, but this is not an infallible rule.
   (a) True
   (b) False

5.5.1.2 Performance Indicators and Contingency Tables

Of course, in most cases, quite sophisticated hypothesis tests could also be performed on the test set, but our aim is mainly to ask whether we can truly separate the groups and how well. Usually, a number of performance indicators are calculated, the most straightforward is the %CC or percentage correctly classified from the test set samples as mentioned above. A poor %CC does not necessarily indicate a poor method.

Sometimes, different classes are predicted with different efficiency, and the data can be presented in the form of a contingency table (also called a confusion matrix). This is illustrated in Table 5.11. The columns represent the known (or diagnosed) classes and the rows represent the predictions. From this table, it can be seen that classes A and B are somewhat less well distinguished compared with class C and are sometimes confused with each other: for example, they may represent two genetically similar organisms, whereas class C is quite distinct genetically.

(a) The per cent correctly classified for class B is 80%.
(b) The per cent correctly classified for class B is 76%.
(c) The per cent correctly classified for class B is 79%.

5.5.1.3 Iterations

When data sets are quite modest in size, however, the performance of a test set may well be influenced by one or two samples, which could, for example, be atypical or ambiguous or outliers; hence, a more sophisticated approach is to generate a test set many times over by selecting different samples in each occasion. Indicators of success can be averaged over each iteration. As computing power becomes more powerful, this repeated validation is less time consuming and, rather than taking hours on a typical desktop or laptop, can be done in minutes or seconds. The calculations can be performed, for example, 100 times over, using different test and training set combinations, and an average or consensus view is obtained.

Usually, the test sets are drawn to represent the same proportion of each class as the overall data set, so that, for example, if a data set of 60 samples consists of 30 each of classes A and B, the test set may consist of 10 samples randomly selected from each class, rather than 20 samples selected from the entire data set, which may be unevenly distributed between classes. This sometimes causes dilemmas when class sizes are unequal, the default is to keep to the original proportions. Hence, if there are 90 samples in class A and 30 in class B, the test set may consist of 30 from class A and 10 from class B, but this is certainly not a universal rule.

There are sometimes algorithms for repeated selection of test set samples, and methods in the literature include double cross-validation and iterative selection of test and training sets.

1. There is a suspected outlier in a data set.
   (a) It is best removed from the data set before validating pattern recognition methods.
   (b) By repeated generation of test and training sets, its influence can be averaged and this is the best approach unless we are certain that we would never encounter such a sample.
   (c) It should always be part of the test set and never the training set.

5.5.1.4 Permutation Methods

There are, however, other quite different approaches. Often, statisticians like to test against a null hypothesis that there is no significant discrimination between classes (in our case). A common approach involving permuting the classifier is the Monte Carlo method.

- The classification of each sample is randomly permuted. Hence, if there are 10 samples of class A and 10 of class B, membership of class A is randomly assigned to any 10 of the original samples and of class B to the other 10. This means that samples will be assigned to either class approximately half the time; obviously, analogously as we cannot guarantee that an unbiased coin will turn up Heads exactly half the time, sometimes as a sample will be chosen as class A slightly more frequently to class B.
- Then, the full classification procedure is performed, often involving a test and training set split, and an indicator of success such as the %CC is calculated on the permuted data, usually using the test set.
- This permutation is repeated many times (e.g. 100 times), each time generating a slightly different value of %CC. These form an ensemble of %CCs for the permuted (basically null) data set.
- The %CC (or any other quantitative indicator of success) of the unpermuted data set is compared with the null ensemble, and if it is much larger, it is considered significant.

The ensemble of results form a background and can be compared with the result on the unpermuted (real) data. Figure 5.17 illustrates typical results of Monte Carlo permutations. The bars represent that the frequency-specific %CC values are obtained, with the vertical line the result for a real data set. It is then possible to put an empirical probability onto this according to where the real result falls on the curve, in our case equivalent to the relative area right of the red line. An advantage of this is that the underlying probability that a sample is correctly classified may be less than 100%; this may be due to overlap, ambiguity, difficulties in finding representative samples or often in metabolomics due to differences between individuals. However, if the test set classification ability is significantly better than that of the null

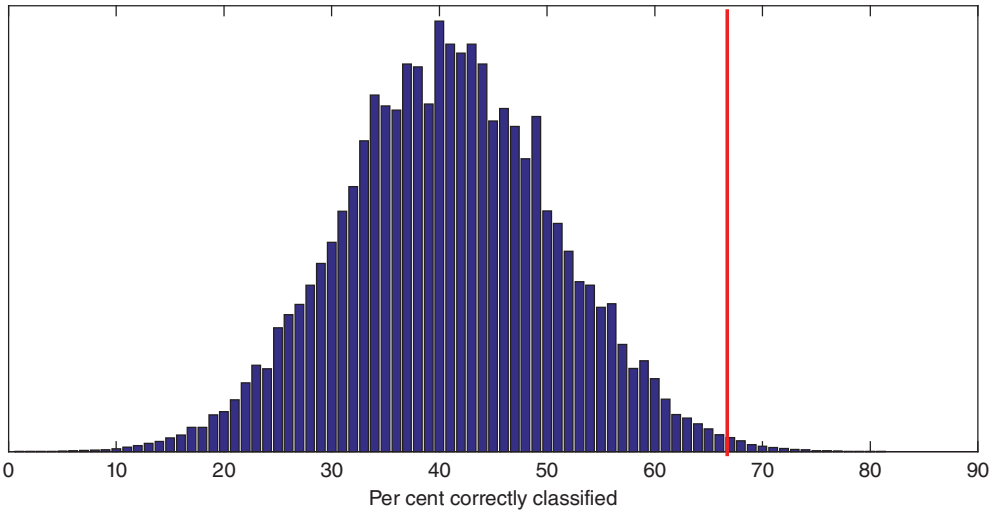


Figure 5.17 Monte Carlo methods: bars represent frequency of results of several permutations for the %CC, whereas the red line represents the unpermuted data.

or permuted data, it suggests that there is a significant difference between groups even if some individual samples are ambiguous.

1. Using PLS-DA, a sample has a *c* value of +1. When performing Monte Carlo methods
   - **(a)** The *c* value is changed to −1.
   - **(b)** The *c* value is changed to +1 or −1 with a 50% chance.
   - **(c)** The *c* value is changed to +1 or −1 many times over, half the time +1 and half the time −1.
   - **(d)** The *c* value is changed to +1 or −1 many times over, approximately half the time +1 and approximately half the time −1.

5.5.2 Optimisation

Optimisation involves obtaining a model that performs as well as possible. In most cases, this involves obtaining a value of %CC as high as possible and choosing the parameters that optimise this value.

In order to avoid overfitting (i.e. obtaining an unduly optimistic view), it is usually performed on the training set. There are many reasons for optimisation. Some of the most common are as follows.

- Choosing the optimum number of principal components.
- Choosing the optimum number of partial least-squares components.
- Choosing the best value of *k* when performing kNN.
- Choosing which variables (e.g. spectroscopic wavelengths or NMR peaks) to retain in the model.
- Reducing the number of variables before LDA.
- Scaling or transforming the data.

The first two are common in chemometrics and can be performed by several methods, some of which have been discussed already in Chapter 4 in the context of PCA. Cross-validation is well known, involving removing single or a group of samples, as a mini-test set, and forming the model on the remaining – the best model has the optimum cross-validated predictions. However, in the case of classification, the predictions are usually of the number of samples correctly classified rather than as a numerical fit to the data; thus, for example, if we had 50 samples in the original data, and found that using one PC or PLS component correctly classified 38 of the cross-validated samples, but two components predicted 42 correctly, and three predicted 46, then the three PC model would be the best of the three. A problem here is that for a single training set, the number correctly classified can be rather flat as the number of components is varied: to overcome this, iterative methods, as discussed in Section 5.5.1.3, can help, generating an average of many different training sets to give an average %CC (as appropriate).

Bootstrap training samples	Bootstrap test samples
10	3
7	5
1	6
9	
8	
7	
4	
1	
2	
9	

Figure 5.18 Typical bootstrap sampling.

An alternative is the bootstrap, which is an iterative method. The main steps in the case of optimising the number of components are as follows.

- The training set is resampled as many times as there are samples.
- Each time, it is sampled repetitively; thus, some samples will be chosen more than once to give a bootstrap training set with some repetitions. Hence, if there are 50 samples in the training set, sample 1 may be chosen two times, sample 2 chosen three times and sample 3 never.
- Then, the ones never chosen are the bootstrap test set.
- Each time, the model is computed on the bootstrap training set: note that this can contain more than one sample of the same kind. It is performed using different numbers of components.
- Its performance is tested on the bootstrap test set for different numbers of components.
- The entire procedure is then repeated many times over, typically 200.
- The bootstrap test set performance for varying number of components is averaged over all these repetitions, and then an optimum number of components is determined.

The bootstrap could be used in other situations, such as optimising k for kNN.
A typical bootstrap sampling is illustrated in Figure 5.18.

- There are 10 samples in the original data set.
- In the bootstrap training set, samples 7, 1 and 9 are selected twice.
- Samples 3, 5 and 6 are never selected in the training set and therefore are part of the bootstrap test set.
- This partition of the data is repeated again many times over, typically 200; thus, the next time we might find samples 2 and 6 are selected twice and sample 9 three times, but samples 1, 4, 5 and 10 are never selected and hence form the bootstrap test set.

Naturally, some samples may be selected more than twice in the training set. The bootstrap test set has certain statistical properties that make this approach optimal.

Figure 5.19 illustrates a typical strategy including the bootstrap; the number of iterations/repetitions, of course, can differ. This also involves regenerating the test set many times over, as recommended in Section 5.5.1.3. Although computationally intense, with modern computer power, it is very feasible in realistic timescales.

There are, of course, other approaches to optimisation and this section is not exhaustive. However, it is important to separate optimisation from validation and also to establish a suitable criterion, usually the per cent correctly classified.

1. A data set consists of ten samples, numbered 1–10. When a bootstrap is used to select samples for the training set, the following 10 samples are selected: 2, 9, 3, 2, 10, 1, 4, 6, 9, 5. The bootstrap test set consists of
   (a) Samples 2 and 9.
   (b) Samples 7 and 8.
   (c) Samples 1, 3, 4, 5 and 6.

5.6 Significant Variables

A final and important topic quite specific to chemometrics relates to which variables are significant. This has an important role especially in metabolic profiling, where we may be interested to know compounds are best at discriminating between groups of samples. For example, which compounds are candidate biomarkers for a specific characteristic?

Table 5.13 Data set mentioned in Section 5.6.

Sample	Class	Variable									
		1	2	3	4	5	6	7	8	9	10
1	A	6.017	3.035	5.004	6.950	3.736	3.155	6.717	5.227	8.592	1.794
2	A	10.022	4.807	9.020	10.083	6.934	4.974	9.659	9.151	13.891	2.436
3	A	9.239	3.333	7.941	8.663	5.848	4.216	7.898	7.094	12.044	2.051
4	A	6.501	3.129	6.712	5.042	5.435	3.011	4.825	6.783	8.742	1.015
5	A	8.036	3.793	7.670	7.245	6.028	3.850	6.912	7.709	10.948	1.639
6	A	6.272	3.368	5.172	7.563	3.871	3.386	7.383	5.588	9.135	1.982
7	A	8.403	5.270	7.937	9.250	6.377	4.599	9.313	9.142	12.521	2.292
8	A	5.763	3.181	4.913	6.768	3.743	3.107	6.639	5.368	8.411	1.749
9	B	5.282	4.683	5.782	7.896	5.814	7.670	6.937	6.305	6.327	2.846
10	B	4.593	3.898	8.999	10.416	6.993	9.232	7.035	4.736	4.601	3.614
11	B	6.926	5.864	9.185	11.508	8.155	11.121	9.389	7.909	7.705	4.054
12	B	6.068	5.144	8.694	10.708	7.503	10.156	8.409	6.818	6.643	3.761
13	B	4.286	3.445	4.956	6.179	4.429	6.443	5.510	4.972	4.695	2.155
14	B	7.442	6.433	9.605	12.291	8.780	11.761	10.087	8.581	8.472	4.361
15	B	4.257	3.315	7.254	8.215	5.532	7.905	6.059	4.499	4.131	2.807
16	B	5.762	4.760	8.193	9.939	6.939	9.621	7.900	6.450	6.181	3.468

Extending this to real situations where we have thousands of variables and often quite limited sample sizes, it is quite possible to turn up potential markers whose distribution could be by chance. There is no really obvious solution to this except to increase the sample size, often by several orders of magnitude, which is usually impractical either in terms of cost (the grant body often has only limited funds), time and even the problem of finding representative samples. Hence, in many situations, the best that can be done is some exploratory analysis.

Nevertheless, chemometrics does play a very important role in pulling out potential markers. In areas such as spectroscopy, it can result in a definitive answer, but in many other areas, it is still also a useful tool for data mining.

In this section, we will illustrate the methods by a simple case study to show the main calculations and their pros and cons, but it is important to understand that the same principles can be extended to far larger real-world data sets. Table 5.13 presents a small 16×10 data set, in which the first eight samples are from class A and the second from class B. We are interested in variables that are most discriminatory. As this should ideally be regarded as an exploratory procedure, it can be done on the auto-predictive or overall data set, rather than a test set.

In this section, we describe three common approaches, but there are, of course, many more described in the literature.

1. An unbiased coin is tossed 12 times. What is the probability that there will be either more than nine or less than three Heads?
   (a) 0.0192
   (b) 0.0385
   (c) 0.0730
   (d) 0.1460

5.6.1 Partial Least-Squares Discriminant Loadings and Weights

Although PLS-DA holds little advantage over traditional statistical methods such as LDA as a classification method, it does hold a significant advantage in exploring variables that are most discriminatory between two or more classes. LDA and most other traditional methods were not designed to provide insight into variables, whereas PLS methods do provide this insight. Many traditional statistical tests use approaches based on the F-statistic, t statistic, ANOVA and so on, which we will discuss in Section 5.6.2, but this neglects interactions or correlations between variables and treat each variable in a univariate manner. Multivariate methods take these correlations into account.

positive scores primarily relate to class A and negative to class B. The two variables with the most positive loadings are 1 and 9, whereas those with the most negative are 6 and 10, being the best markers for classes A and B respectively. Note that using most PLS1 algorithms, PLS loadings are not normalised, in contrast to PCA, or centred. For a rule of thumb, we select the extreme variables, which we visually identify as the markers for each group. The origin is not always significant, as the centre of gravity may lie one side of 0. Some workers prefer weights to loadings, as the latter are orthogonal; however, if the conclusions are visual, there is rarely a major difference in most practical situations.

There is a vast literature interpreting PLS-DA loadings, but the aim of this section is to introduce the usage as a method for determining variables that correspond to potential markers. PLS-DA is a very useful exploratory visual method, which can also be used to rank variables according to their significance.

1. PLS1-DA loadings for individual components where there are two classes usually
   (a) Have zero mean and a sum of squares of 1.
   (b) Have zero mean and a mean sum of squares of 1.
   (c) Have zero mean but do not necessarily have a sum (or mean sum) of squares of 1.
   (d) Do not necessarily have a zero mean or a sum (or mean sum) of squares of 1.

5.6.2 Univariate Statistical Indicators

It is also possible to determine univariate statistical indicators of whether variables are significant or not. There are various ways but they are mostly related. These can be done on the raw data.

One of the simplest is the t statistic defined as

$$t_j = (\bar{x}_{j\mathrm{A}} - \bar{x}_{j\mathrm{B}}) \Big/ (s_{jpooled} \sqrt{(1/I_{\mathrm{A}} + 1/I_{\mathrm{B}})})$$

In our case for variable 3,

- $\bar{x}_{3\mathrm{A}} = 3.740$ or the mean of the eight values for class A
- $\bar{x}_{3\mathrm{B}} = 5.578$
- $s_{3pooled} = (s_{3\mathrm{A}} + s_{3\mathrm{B}})/2$ as both groups are equal sized or $(1.488 + 1.582)/2 = 1.535$
- The term $\sqrt{(1/I_{\mathrm{A}} + 1/I_{\mathrm{B}})}$ is not strictly necessary if we are just interested in ranking variables, but it essential if the t value is to be converted to a probability, and in our case equals 0.5.
- Hence, the t value for variable is $(3.740 - 5.578)/(1.535 \times 0.5)$ or -1.352.
- This value could be converted into a p value using the t statistic, but this depends on the variables being normally distributed within each class, a condition that is rarely met. Although, it can also just be used to numerically rank and assign variables.

The t values for our data set are presented in Figure 5.22. For PLS-DA, we can calculate the loadings and weights for each PLS component, but for straightforward univariate statistics, there is only one value for each variable. In this example, we see that variables 6 and 10 are most diagnostic for class B and 1 and 9 for class A, a similar conclusion to PLS-DA; however by no means, all situations will be in agreement between the methods.

The univariate statistical tests are valuable when several factors result in separation. Although we may be able to separate two classes, say Male and Female, there may be other factors such as Young and Old and so the data could be grouped in different ways. There are elaborate extensions such as various types of ANOVA and multi-level approaches, often used in clinically designed experiments. There are also methods of combining multivariate methods with multi-level univariate approaches such as ASCA. However, these are outside the scope of a basic introduction and will only be necessary in certain specialist cases, normally involving designed experiments such as in clinical laboratories. In order to analyse such data, quite a specialist expertise is required.

1. Class A consists of 50 samples and class B of 100. For one of the variables, the mean for class A is 3.29 and class B 5.71. The standard deviation over class A is 1.28 and class B is 2.79. What is the value of t, assuming class A is positive?

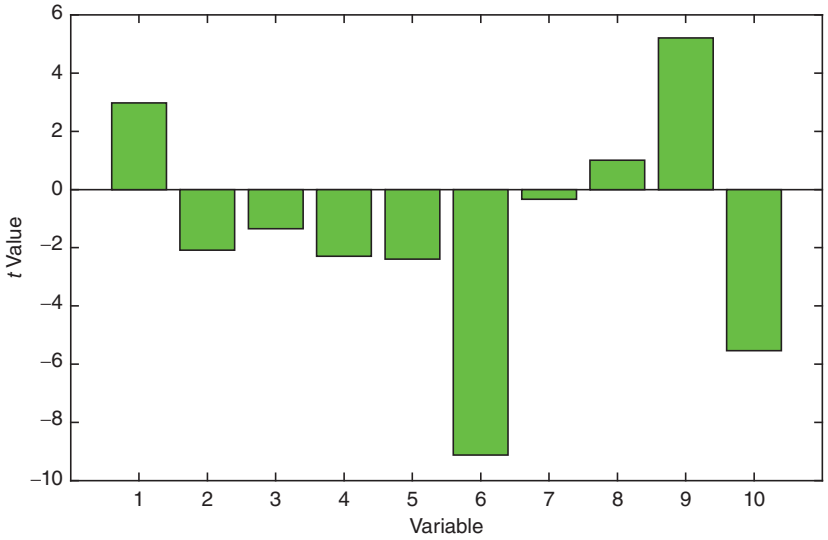


Figure 5.22 Values of *t* for the 10 variables in Table 5.13.

(a) −1.057
(b) −0.183
(c) −1.189
(d) −0.206

5.6.3 Variable Selection for SIMCA

The methods discussed in the last two sections are for two (or multi)-class classifiers. Strictly speaking, as SIMCA models each class separately, we should not be asking whether we can find variables that discriminate between classes. However, the classical SIMCA literature does define discriminatory power, so we report this for completion.

In order to determine this, it is necessary to fit each sample to both class (disjoint PC) models. For example, fit sample 1 to the PC model of both class A and class B. The residual matrices are then calculated, but there are now four such matrices.

- Samples in class A fitted to the model of class A.
- Samples in class A fitted to the model of class B.
- Samples in class B fitted to the model of class B.
- Samples in class B fitted to the model of class A.

We would expect matrices 2 and 4 to be a worse fit than matrices 1 and 3. The standard deviations for each variable are then calculated for these matrices to give

$$D_j = \sqrt{\frac{\text{class A}\quad \text{model B}\, s^2_{jresid} + \text{class B}\quad \text{modelA}\, s^2_{jresid}}{\text{class A}\quad \text{model A}\, s^2_{jresid} + \text{class B}\quad \text{model B}\, s^2_{jresid}}}$$

The bigger the value, the higher the discriminatory power and the more significant the variable. Discriminatory power can of course be calculated between any two classes.

1. Two groups are modelled. The standard deviation of variable 7 class A to class A is 12.35 and to class B is 17.12, whereas variable 7 class B to class A is 21.51 and to class B is 13.56. The discriminatory power is
   (a) 1.50
   (b) 0.67
   (c) 0.83